

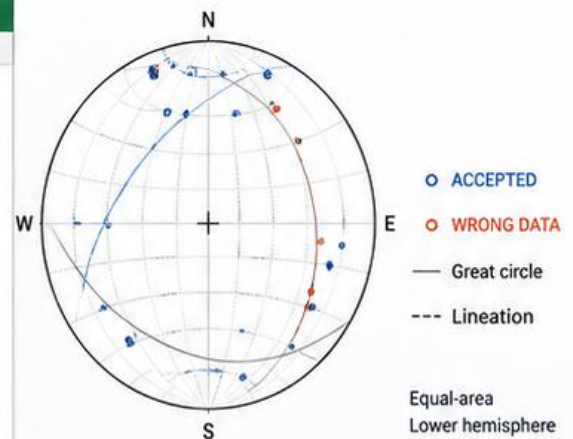


StruTest 4.0

A Quantitative Quality Assessment Tool for Structural Datasets in Digital Geological Mapping

— Based on Excel Tables —

StruTest 4.0										
File Home Insert Page Layout Formulas Data Review View Help										
Test of Linear										
ID	Input Data					Calculated Data		Errors (Measured - Calculated)		Quality
	Strike (°)	Dip Dir. (°)	Dip (°)	Plunge Dir. (°)	Plunge (°)	β (°)	Calc. Plunge (°)	Plunge Angle Error (°)	Plunge Dir. Error (°)	Result
1	135	225	68	210	32	75	31.4	0.6	5.0	ACCEPTED
2	78	168	45	120	20	42	19.1	0.9	11.0	ACCEPTED
3	210	300	82	250	15	50	14.3	0.7	10.0	ACCEPTED
4	350	80	30	60	8	30	7.6	0.4	0.0	ACCEPTED
5	125	215	60	190	35	75	34.1	0.9	5.0	ACCEPTED
...	...	...	...	...	...	...	...	...	...	...
254	48	138	55	120	28	57	27.0	1.0	0.0	ACCEPTED
255	302	32	25	310	5	28	4.8	0.2	8.0	WRONG DATA



Data Validation

Automatic detection of inconsistent records



Error Estimation

Quantitative plunge angle and plunge direction errors



Quality Summary

Statistics and quality rating of the entire dataset



Adjustable Thresholds

User-defined acceptance criteria



Excel-based

Easy to use and highly compatible



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User Manual for StruTest4.0

1. Introduction

StruTest4.0 is an Excel-based software developed for the quantitative quality assessment of structural orientation datasets collected during digital geological mapping and other structural studies. It provides an efficient and user-friendly workflow for validating planar and linear structural measurements according to their fundamental geometric relationships and for estimating measurement errors.

For planar measurements, the **StruTest4.0** automatically verifies the consistency between strike (S) and dip direction (D) based on the British Right-Hand Rule ([Figure 1a](#)) and identifies invalid measurements. For linear measurements, the software first checks whether the associated planar measurement is valid and then evaluates the geometric relationship between the lineation and its hosting plane. It further calculates errors of plunge angle (Pa) and plunge direction (P) ([Figure 1b](#)) and classifies each measurement according to user-defined quality threshold.

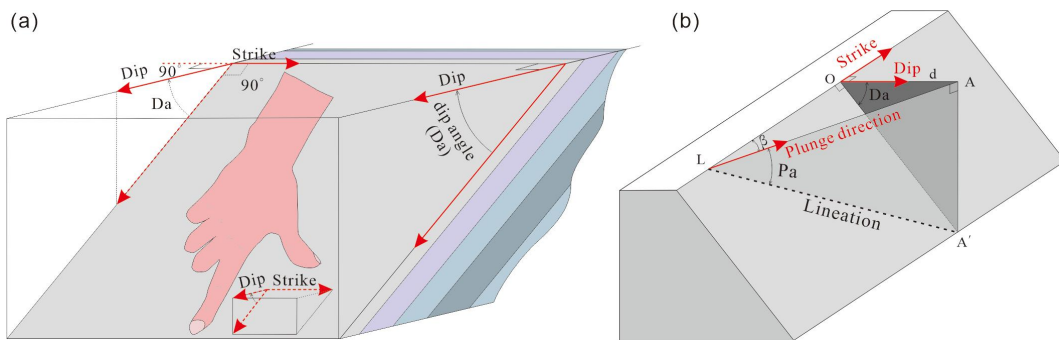


Figure 1 Cartoons showing structural attitudes for a planar and a linear structures using the British Right-hand Rule to indicate the correlation between strike and dip for an inclined plane (a), and to show the spatial correlation between an inclined lineation within an inclined plane (b). Da: dip angle, Pa, plunge angle.

Compared with conventional quality control methods based on stereographic projection, the **StruTest4.0** performs automatic validation and quantitative error estimation for large structural datasets. It provides detailed comments for each measurement together with summary statistics, enabling rapid identification of incorrect measurements and evaluation of the overall quality of structural datasets.

This user manual describes the software requirements, input data format, workflow, output interpretation, and example applications to help users efficiently

perform quality assessment using **StruTest4.0**.

2. System Requirements

StruTest4.0 is developed as a Microsoft Excel workbook and does not require installation or additional software packages. The following system configuration is recommended to ensure normal operation ([Table 1](#)).

Table 1 Requirement list of the StruTest4.0

Item	Requirement
Operating system	Windows 10 or later (Windows 11 recommended)
Software	Microsoft Excel 2016 or later (Microsoft 365 is fully supported)
File format	Microsoft Excel Workbook (*.xlsx)
Additional software	None
Internet connection	Not required for software operation

Notes:

- StruTest4.0 is distributed as a standard Microsoft Excel workbook (.xlsx) and can be opened directly in Microsoft Excel.
- The software does not require installation, compilation, programming environments, or third-party libraries.
- To ensure full compatibility, it is recommended to use Microsoft Excel on the Windows platform. Although the workbook may open on other operating systems, complete functionality has not been systematically tested.

3. Software Overview

StruTest4.0 is implemented as a Microsoft Excel workbook and consists of two main worksheets designed for the quality assessment of planar and linear structural measurements ([Figure 2](#)). The software provides an intuitive workflow that enables users to import structural orientation data, perform automatic validation, estimate orientation errors, and obtain an overall quality evaluation of the dataset.

The two worksheets are summarized below.

Test of Plane

This worksheet is used to evaluate planar structural measurements. It automatically checks the consistency between strike (S) and dip direction (D) according to the British Right-Hand Rule, identifies invalid measurements, and summarizes the overall quality of the planar dataset.

Test of Linear

- This worksheet is designed for evaluating linear structural measurements together with their corresponding planar orientations. The software verifies the geometric relationships between lineations and their hosting planes, calculates plunge angle and plunge direction errors, classifies each measurement, and provides an overall quality evaluation of the linear dataset.

The software performs all calculations automatically once the structural measurements are entered into the corresponding worksheets. No additional configuration or programming is required.

Data Sequence	Paste Measured Data Below				TEST	
	Strike	D_angle	Dip	D-S	Comments	
	S	Da	D			
1	212	30	302	90	ACCEPTED	
2	175	30	265	90	ACCEPTED	
3	265	70	355	90	ACCEPTED	
4	250	70	340	90	ACCEPTED	
5	204	46	294	90	ACCEPTED	
6	267	73	357	90	ACCEPTED	
7	201	23	291	90	ACCEPTED	
8	235	42	325	90	ACCEPTED	
9	201	28	291	90	ACCEPTED	
10	190	28	280	90	ACCEPTED	
11	111	89	201	90	ACCEPTED	
12	340	66	70	90	ACCEPTED	
13	274	82	4	90	ACCEPTED	
14	220	24	310	90	ACCEPTED	
15	158	23	248	90	ACCEPTED	
16	145	30	235	90	ACCEPTED	
17	56	62	146	90	ACCEPTED	
18	135	40	225	90	ACCEPTED	
19	173	35	263	90	ACCEPTED	
20	260	45	350	90	ACCEPTED	
21	165	25	255	90	ACCEPTED	
22	111	16	201	90	ACCEPTED	
23	255	78	345	90	ACCEPTED	
24	125	79	215	90	ACCEPTED	

Test of Plane

Test of Lineation

+

(a)

Data Sequence	Paste Measured Data Below					Calculated		Estimated		Error				Consistency of the Measured Data	
	Strike	D_angle	Dip	Plunge	P_angle	D(S-D-S)	β (P*S)	Plunge (Da-Pa)	$\tan P \tan Da \sin \beta$	Plunge angle	Comments P_angle	Plunge direction	Comments Plunge		
	S	Da	D	P	Pa				P_angle	β (P*S)					
1	175	30	265	333	12	90	158	18	12	158	0	Accepted	0	Accepted	ACCEPTED
2	265	70	355	275	27	90	10	43	28	11	1	Accepted	1	Accepted	ACCEPTED
3	190	28	280	345	14	90	155	14	13	152	1	Accepted	3	Accepted	ACCEPTED
4	274	82	4	277	20	90	3	62	20	3	0	Accepted	0	Accepted	ACCEPTED
5	111	16	201	174	14	90	63	2	14	60	0	Accepted	3	Accepted	ACCEPTED
6	125	79	215	292	50	90	167	29	49	167	1	Accepted	0	Accepted	ACCEPTED
7	85	75	175	238	57	90	153	18	59	158	2	Accepted	3	Accepted	ACCEPTED
8	185	32	275	185	2	90	0	30	0	3	2	Accepted	3	Accepted	ACCEPTED
9	170	30	260	181	9	90	11	21	6	16	3	Accepted	5	Accepted	ACCEPTED
10	133	39	223	167	23	90	34	16	24	32	1	Accepted	2	Accepted	ACCEPTED
11	183	39	273	357	6	90	174	33	5	173	1	Accepted	1	Accepted	ACCEPTED
12	197	40	287	3	11	90	166	28	11	167	0	Accepted	1	Accepted	ACCEPTED
13	199	20	289	356	10	90	157	10	8	151	2	Accepted	6	Error	ACCEPTED
14	184	20	274	341	8	90	157	12	8	157	0	Accepted	0	Accepted	ACCEPTED
15	164	46	254	192	26	90	28	20	28	28	0	Accepted	0	Accepted	ACCEPTED
16	152	48	242	175	24	90	23	24	23	24	1	Accepted	1	Accepted	ACCEPTED
17	174	64	264	185	19	90	11	45	21	10	2	Accepted	1	Accepted	ACCEPTED

Test of Plane

Test of Lineation

+

(b)

Figure 2 Main interface of StruTest4.0 showing the two worksheets of *Test of Plane* (a) and *Test of linear* (b)

4. Input Data

StruTest4.0 accepts structural orientation measurements in numeric format. Before performing quality assessment, users should enter the structural measurements into the corresponding worksheets. The software automatically performs validation and error estimation after the required parameters have been entered.

4.1 Planar Measurements

Planar structural measurements should be entered in the **green input cells** of the *Test of Plane* worksheet (Figure 2a). Users only need to input the required orientation parameters into the designated green area, whereas all other cells contain formulas or automatically generated results and should not be modified. Each record consists of three orientation parameters:

- Strike (S, °)
- Dip angle (Da, °)
- Dip (D, °)

All orientation values should be expressed in degrees. Strike and dip direction should follow the British Right-Hand Rule (Figure 1). The software automatically checks the consistency between strike and dip (dip direction) and identifies measurements that violate the required geometric relationship.

Table 2 Required input parameters for planar measurements

Item	Requirement	Unit
Strike	Strike of the planar structure	Degree (°)
Dip angle	Dip angle of the planar structure	Degree (°)
Dip	Dip direction of the planar structure	Degree (°)

4.2 Linear Measurements

Linear structural measurements should be entered in the **green input cells** of the *Test of Linear* worksheet (Figure 2b). Users only need to input the required orientation parameters into the designated green area, whereas all calculation results, comments, and quality assessment fields are generated automatically and should not be edited.

The required input parameters include:

- Plunge (Plunge direction/Trend) (P, °)
- Plunge angle (Pa, °)
- Corresponding planar orientation (see *Section 4.1*), include:
 - Strike (S, °)
 - Dip angle (Da, °)
 - Dip direction (D, °)

Table 3 Required input parameters for linear measurements

Item	Requirement	Unit
Plunge	Trend of the lineation	Degree (°)
Plunge angle	Plunge angle of the lineation	Degree (°)
Strike	Strike of the planar structure	Degree (°)
Dip angle	Dip angle of the planar structure	Degree (°)
Dip	Dip direction of the planar structure	Degree (°)

The software first verifies the validity of the corresponding planar measurement and then evaluates the geometric relationship between the lineation and its hosting plane. Measurements that do not satisfy the required geometric relationships are identified automatically.

Note:

- If some planar measurements are not associated with lineation measurements, it is recommended to use the sorting function in Microsoft Excel to separate records with valid lineation data from those containing blank values. Only records with complete lineation measurements need to be copied into the ***Test of Linear*** worksheet. Removing them helps minimize unnecessary interference during the evaluation process and improves the efficiency of data processing.

4.3 Data Requirements

To ensure accurate quality assessment, users should follow these recommendations:

- All input values must be numeric.
- Each row should represent one structural measurement.
- Linear measurements must be entered together with their corresponding planar measurements.

5. Workflow

The workflow of **StruTest4.0** is straightforward and requires only a few steps to complete the quality assessment of structural orientation datasets. The recommended procedure is illustrated in [Figure 3](#) and described below.

Step 1. Prepare the structural dataset

Prepare the structural orientation measurements in Microsoft Excel. Planar measurements should include **strike, dip angle, and dip (dip direction)**, whereas linear measurements should include **plunge (plunge direction), plunge angle**, and the corresponding planar orientation. Before copying the data into StruTest4.0, verify that the order of the input columns is consistent with that required by the software.

Step 2. Evaluate planar measurements

Copy the planar measurements into the Test of Plane worksheet. The software

automatically validates the geometric relationship between strike and dip direction according to the British Right-Hand Rule and summarizes the quality of the planar dataset.

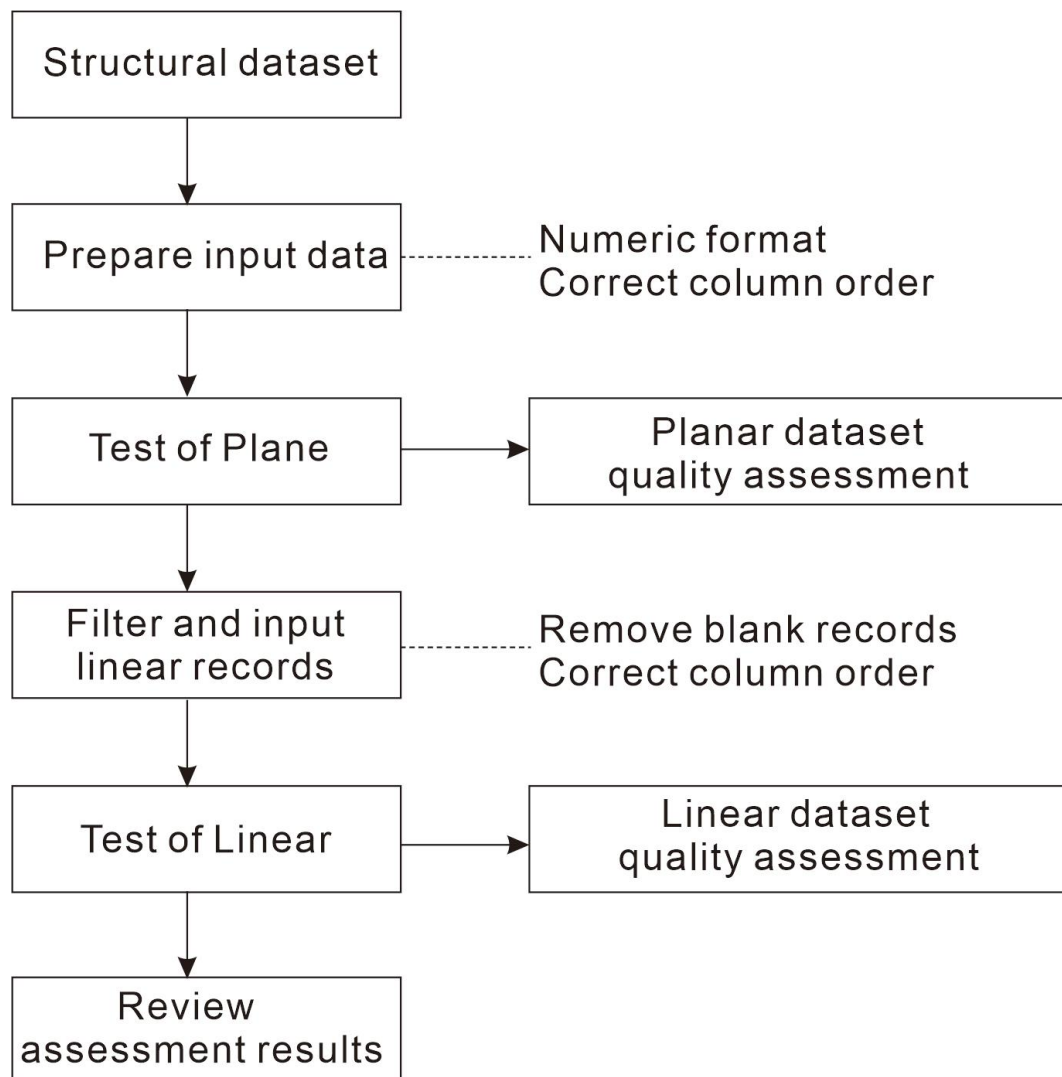


Figure 3 Recommended workflow for quality assessment using StruTest4.0.

Step 3. Prepare linear measurements

If linear measurements are available, sort the structural dataset using the sorting function in Microsoft Excel and copy only the records containing valid lineation measurements into the Test of Linear worksheet. Excluding the blank record simplifies the evaluation process and improves operational efficiency.

Step 4. Evaluate linear measurements

After the linear measurements are entered, StruTest4.0 automatically verifies the geometric relationships between the lineation and its corresponding planar orientation,

calculates plunge angle and plunge direction errors, classifies each measurement, and summarizes the overall quality of the linear dataset.

Step 5. Review the assessment results

The assessment results are generated automatically as planar and linear measurements are entered into the corresponding worksheets. Review the validation results, comments, orientation errors, and quality statistics generated by the software. Measurements classified as **WRONG DATA** (of planar and linear measurements) should be corrected or discarded. Measurements classified as **ERROR** for linear measurements satisfy the required geometric relationships but exceed the user-defined acceptable error threshold and should therefore be examined carefully before further analysis.

6. Output Interpretation

6.1 Output of the *Test of Plane* worksheet

The *Test of Plane* worksheet provides 10,000 rows for planar measurement input and automatic output, which is sufficient for most geological mapping projects. For larger datasets, users can extend the worksheet by copying the existing input and output rows below the predefined range. This allows **StruTest4.0** to accommodate datasets containing more than 10,000 measurements without affecting its functionality. Each input measurement is evaluated automatically as soon as the required orientation parameters are entered.

For each planar measurement, the software generates the following outputs:

- **Validation result** of the geometric relationship between strike and dip direction (D–S check)
- **Comments** describing the validation result for each measurement

In addition, the worksheet summarizes the quality of the entire planar dataset, including:

- **Total** number of input measurements
- **Number** and percentage of **ACCEPTED** measurements
- **Overall quality evaluation** of the dataset, classified as **High-quality, Medium-quality, or Low-quality**

The overall quality evaluation is calculated automatically based on the

proportion of **ACCEPTED** measurements. By default, datasets with more than **99%** accepted measurements are classified as **High-quality**, those with **95 – 99%** accepted measurements as **Moderate-quality**, and those with less than **95%** accepted measurements as **Low-quality**. These classification thresholds are user-configurable and may be modified to meet the quality assessment criteria of specific projects.

	Input in green cells only			Output automatically		Evaluation Quality of Planar Orientations			
Data Sequence	Paste Measured Data Below			TEST					
	Strike	D_angle	Dip	D-S	Comments	Total	Accepted	Percentage	Evaluation
	S	Da	D			6875	6865	99.85	High-Quality
1	212	30	302	90	ACCEPTED				
2	175	30	265	90	ACCEPTED				
3	265	70	355	90	ACCEPTED				
4	250	70	340	90	ACCEPTED				
5	204	46	294	90	ACCEPTED				
6	267	73	357	90	ACCEPTED				
7	201	23	291	90	ACCEPTED				
8	235	42	325	90	ACCEPTED				

Figure 4 Input and Output of the *Test of Plane* worksheet.

6.2 Output of the Test of Linear worksheet

The *Test of Linear* worksheet also provides 10,000 rows for planar measurement input and automatic output. This allows **StruTest4.0** to accommodate datasets containing more than 10,000 measurements without affecting its functionality. Each input measurement is evaluated automatically as soon as the required orientation parameters are entered.

The output of the *Test of Linear* worksheet consists of four categories: **validation results**, **calculated values**, **measurement evaluation**, and **dataset evaluation**.

(1) Validation results

The software automatically verifies the geometric relationships between the lineation and its corresponding planar measurement, including:

- **Validation of the strike–dip direction relationship (D–S check)** of the associated planar measurement;
- **Validation of the included angle (β)** between the strike (S) of the planar measurement and the plunge direction (P) of the lineation;
- **Validation of the relationship between the dip angle (Da)** of the planar measurement and the **plunge angle (Pa)** of the lineation.

(2) Calculated values

For each valid linear measurement, StruTest4.0 automatically calculates:

- The theoretical plunge angle;
- The theoretical included angle (β) between the plunge direction and the strike;
- The plunge angle error;
- The plunge direction error (difference between the measured and theoretical β values).

The calculated plunge angle and plunge direction errors are compared with the corresponding acceptable error thresholds to determine whether they fall within the allowable range.

(3) Measurement evaluation

Each linear measurement is automatically classified as one of the following:

- **WRONG DATA** – the measurement violates the required geometric relationships and should be corrected or discarded;
- **ERROR** – the geometric relationships are valid, but both plunge angle and plunge direction errors exceed the acceptable threshold;
- **ACCEPTED** – the measurement satisfies both the geometric relationships and the user-defined error criteria.

The acceptable error thresholds for plunge angle and plunge direction are determined automatically according to the geometric characteristics of the measurement, such as the dip angle of the hosting plane and the spatial relationship between the lineation and the planar measurement. Users may also modify these thresholds to meet the quality-control requirements of different projects.

(4) Dataset evaluation

The worksheet automatically summarizes the quality of the entire linear dataset, including:

- **Total** number of linear measurements;
- Number and percentage (W%) of **WRONG DATA** records;
- Number and percentage (A%) of **ACCEPTED** measurements;
- Overall quality evaluation, classified as **High-quality** ($W\% < 1\%$ and $A\% \geq 95\%$), **Moderate-quality** (all other cases), or **Low-quality** ($W\% \geq 3\%$ or $A\% \leq 90\%$).

The overall quality evaluation is based on both the proportion of **WRONG DATA** records and the proportion of **ACCEPTED** measurements, providing a rapid assessment of the reliability of the linear structural dataset. The classification thresholds are user-configurable and can be modified to meet the quality-control requirements of different projects.

Output calculated values automatically					Output theoretical errors (δ) and comments for each measurement								Evaluation Quality of Linear Orientations					
Test		Calculated			Error					Comments on the Measured Data	Threshold		Total	Wrong data	Percentage	Accepted	Percentage	Evaluation
D ^a S (D-S)	β (P ^a S)	Plunge (Da-Pa)	tanPa=tanDa ^a sinβ		Plunge angle	Comments P_angle	Plunge direction	Comments Plunge	can be modified									
			P_angle	β (P ^a S)	δ		δ		Pa (to Da)		P (to β)							
90	2	37	2	5	2	Accepted	3	Accepted	ACCEPTED	3	5							
90	32	10	11	30	0	Accepted	2	Accepted	ACCEPTED	2	3							
90	161	20	10	165	2	Accepted	4	Accepted	ACCEPTED	2	5							
90	173	38	7	172	1	Accepted	1	Accepted	ACCEPTED	4	5							
90	172	63	17	179	15	Error	7	Error	ERROR	5	5							
90	165	45	17	176	12	Error	11	Error	ERROR	4	5							
90	172	50	11	176	6	Error	4	Accepted	ACCEPTED	4	5							
140	227	51	-51	WRONG DATA	59	Error	Error	Error	WRONG DATA	4	5							
90	34	12	57	36	1	Accepted	2	Accepted	ACCEPTED	5	3							

Figure 5 Output of the *Test of linear* worksheet.

7. Example application

7.1 Geological mapping dataset

A structural dataset used to demonstrate the workflow of **StruTest4.0** is provided in the **Example_Data**. It was obtained from the field database of a digital geological mapping project conducted by the China Geological Survey (CGS). This dataset contains 6,875 planar structure measurements, including 255 lineation measurements associated with their corresponding planar structures.

In addition to structural orientation data, the dataset includes the structural types of both planar and linear features, brief field descriptions, and the kinematic interpretations of the lineations based on field observations.

7.2 Quality assessment process

This example demonstrates the complete workflow for assessing the quality of a structural dataset using **StruTest4.0**. The assessment consists of the following five steps:

Step 1. Open the **Example_Data** worksheet and convert all measurement values to **Numeric** format before starting the quality assessment.

Step 2. Copy the planar measurements (strike, dip angle, and dip direction) into

the *Test of Plane* worksheet. The worksheet automatically checks the geometric relationship between strike and dip direction and identifies invalid measurements.

Step 3. Use the Sort function in Microsoft Excel to sort the dataset by the Plunge Direction column. This automatically groups the blank rows together, allowing them to be removed easily. After deleting the blank rows, 255 records containing lineation measurements remain. Copy the corresponding planar measurements together with the associated lineation measurements (plunge direction and plunge angle) into the *Test of Linear* worksheet.

Step 4. The worksheet automatically verifies the geometric relationship between each lineation and its corresponding planar structure, calculates the theoretical plunge angle, estimates the plunge angle and plunge direction errors, and evaluates the quality of each measurement.

Step 5. Review the summary statistics and quality assessment results generated automatically for both the planar and linear structural datasets.

7.3 Interpretation of results

After the example dataset is processed, StruTest 4.0 automatically generates the quality assessment results for both the planar and linear structural datasets.

(1) Planar measures

For the **6,875** planar structure measurements, **10** records are identified as **WRONG DATA**. Examination of these records shows that the angle between strike and dip direction is not 90°, violating the geometric relationship required by the right-hand rule. Therefore, these records are correctly classified as **WRONG DATA**.

The remaining **6,865** measurements satisfy the required geometric relationship and are classified as **ACCEPTED**, corresponding to an acceptance rate of 99.85%. According to the predefined quality criteria, the planar structural dataset is therefore classified as **High-quality**.

(2) Linear measures

For the 255 lineation measurements, one record is classified as **WRONG DATA** because its associated planar measurement is invalid (the angle between strike and dip direction is not 90°). In addition, the angle (β) between the plunge direction and the

strike of the associated plane is 227°, which is outside the valid range of 0°–180°. Therefore, this record is also correctly identified as **WRONG DATA**.

Two measurements are classified as **ERROR** because their plunge angle errors are 15° and 12°, respectively, while their plunge direction errors are 7° and 11°, both exceeding the predefined acceptance thresholds. These records represent measurements with relatively large orientation errors.

In addition, 21 measurements show relatively large plunge direction errors (6–26°) but very small plunge angle errors ($\leq 3^\circ$), and two (2) additional measurements exhibit relatively large plunge angle errors (3–6°) but only small plunge direction errors (4–5°). Although one of the error components is relatively large, the overall orientation errors remain within the acceptable range. Consequently, these measurements are classified as **ACCEPTED**.

Inspection using stereographic projection shows that the lineations are plotted close to the great circles of their corresponding planes, indicating that these measurements are geometrically reasonable. The stereographic projection results are consistent with the automated assessment produced by **StruTest 4.0**.

8. Notes and Limitations

Interpretation of measurements with relatively large errors

The calculated plunge angle and plunge direction errors are affected by both the measured plunge direction and dip angle. For some geometries, a small error in plunge angle may produce a relatively large theoretical plunge direction error, whereas a small deviation in plunge direction may similarly amplify the theoretical plunge angle error. Consequently, some geometrically reasonable measurements may occasionally be classified as **ERROR** and should be examined further rather than being discarded directly.

To accommodate different geological settings and measurement accuracies, **StruTest 4.0** provides adjustable acceptance thresholds. By default, the acceptable plunge angle error is 2°, 3°, 4°, 5°, and 6° for planes with dip angles of (0–30°, 30–45°, 45–60°, 60–75°, and 75–90°, respectively. The default plunge direction error threshold is 7° when β falls within 0°–30° or 150°–180°, and 5° when β falls within 30°–150°. Users may modify these thresholds according to the quality-control requirements and measurement accuracy of their projects, thereby reducing the

unnecessary classification of geometrically reasonable measurements as **ERROR**.

For measurements that remain difficult to interpret after threshold adjustment, stereographic projection is recommended as an additional means of verification. By combining the automated assessment with stereographic analysis, users can make a more reliable judgment of structurally complex or ambiguous measurements.

9. Citation

If you use StruTest 4.0 in academic research, please cite the following publication:

Zha, X., Zhang, Y., Yan, J., Li, R., Qubsani, A., Guo, L., Liu, J., and Xu, L., 2026. A semi-automated quality control framework for structural orientation datasets in digital geological mapping. *Computers & Geosciences*, submitted.